

PISA GALILEO GALILEI, Italy

WMO: 161580

Lat: 43.6823N

Lon: 10.3955E

Elev: 20

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.3	30.5	12.7	10.5	40.0	17.5	13.3	40.5	25.1	52.5	22.5	50.8	5.4	110	0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.6	89.9	71.2	87.9	71.2	86.0	70.8	76.7	83.4	75.2	82.1	73.9	81.0	9.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.0	131.1	80.2	73.3	123.7	79.2	71.5	116.5	77.9	40.1	83.8	38.6	82.2	37.3	80.9	90.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.3	18.8	16.2	DB	23.8	95.0	2.4	2.6	22.0	96.9	20.6	98.4	19.3	99.9	17.6	101.8	(4)
(5)			WB	22.7	79.6	2.4	2.9	20.9	81.7	19.5	83.4	18.1	85.0	16.4	87.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.3	44.8	45.9	50.6	55.6	62.5	70.0	75.0	75.5	68.6	61.8	53.5	46.4	(6)	
	DBStd	11.86	5.33	5.14	4.43	4.28	4.05	4.75	3.63	3.71	4.43	4.97	6.20	6.11	(7)	
	HDD50	539	176	129	46	6	0	0	0	0	0	1	37	144	(8)	
	HDD65	3059	626	535	448	282	99	9	0	0	16	123	345	576	(9)	
	CDD50	3920	14	14	64	174	388	601	774	791	559	366	142	33	(10)	
	CDD65	967	0	0	0	0	22	160	309	327	125	23	1	0	(11)	
	CDH74	8414	0	0	0	6	198	1420	2979	2968	782	61	0	0	(12)	
	CDH80	2701	0	0	0	0	23	412	1053	1068	144	1	0	0	(13)	
(14) Wind	WSAvg	6.4	6.6	6.7	7.0	6.6	6.0	6.0	6.2	5.9	6.2	6.3	7.0	6.9	(14)	
(15) Precipitation	PrecAvg	34.8	2.9	2.7	2.8	2.9	2.4	1.7	0.9	2.0	3.6	5.0	5.0	3.4	(15)	
	PrecMax	57.9	9.4	8.2	9.5	9.9	7.1	4.6	3.3	9.7	12.6	15.1	14.7	7.8	(16)	
	PrecMin	20.1	0.0	0.1	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.2	0.0	0.0	(17)	
	PrecStd	7.7	2.0	1.7	2.0	2.0	1.9	1.1	0.9	1.9	2.5	3.7	3.0	1.8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.4	61.0	67.9	74.8	82.7	91.3	93.0	94.3	87.4	78.7	69.4	62.3	(19)	
		MCWB	53.3	51.3	54.6	60.6	65.6	73.1	70.5	70.7	68.2	64.6	63.0	57.0	(20)	
	2%	DB	57.4	58.8	64.1	71.3	78.9	87.8	89.8	90.1	84.1	75.5	66.5	59.4	(21)	
		MCWB	52.7	50.9	53.1	58.3	63.9	71.3	70.4	71.6	68.1	65.3	60.7	55.0	(22)	
	5%	DB	55.7	57.0	61.9	68.0	75.5	84.6	87.8	87.9	80.9	73.2	64.5	57.6	(23)	
		MCWB	50.9	50.6	52.8	57.3	63.0	69.5	71.0	71.6	67.7	64.9	59.0	53.4	(24)	
	10%	DB	53.8	55.3	59.3	65.0	73.1	82.1	85.9	85.9	78.7	70.9	62.5	55.7	(25)	
		MCWB	49.4	49.7	51.8	56.3	62.1	69.0	70.8	71.4	66.9	63.4	58.0	51.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.7	55.4	58.3	63.1	68.9	77.8	78.0	78.9	74.6	70.4	64.9	58.5	(27)	
		MCDB	58.4	57.7	62.8	69.7	77.1	85.3	84.3	84.2	80.5	73.9	67.6	60.8	(28)	
	2%	WB	53.8	53.5	56.1	60.8	66.7	74.6	76.3	77.1	72.6	68.5	62.0	56.5	(29)	
		MCDB	56.8	56.6	60.6	67.2	74.9	82.7	83.1	83.6	78.3	72.3	65.3	59.0	(30)	
	5%	WB	52.2	52.0	54.9	59.2	65.2	72.4	74.8	75.6	70.9	66.7	60.2	54.7	(31)	
		MCDB	55.2	55.4	59.2	65.4	72.8	80.5	82.1	82.1	77.1	70.8	63.4	57.2	(32)	
	10%	WB	50.5	50.6	53.6	57.5	63.6	70.5	73.6	74.1	69.2	64.9	58.6	52.7	(33)	
		MCDB	53.0	53.8	57.9	63.3	70.5	78.4	81.0	81.0	75.6	69.0	61.8	55.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	14.3	15.7	16.8	17.4	18.3	19.5	20.1	19.6	18.5	16.3	13.4	13.6	(35)	
		MCDBR	14.7	18.1	20.7	21.9	23.5	23.4	23.2	22.2	21.4	18.2	14.0	13.4	(36)	
		MCWBR	10.7	12.1	12.5	12.0	11.7	10.7	10.0	10.0	10.0	10.0	9.0	9.6	(37)	
	5% WB	MCDBR	12.4	14.2	16.7	18.9	20.2	20.8	19.7	19.3	17.7	14.4	11.9	10.4	(38)	
		MCWBR	10.0	11.0	11.4	11.7	11.4	11.0	10.2	10.3	10.0	9.6	8.8	8.6	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.336	0.356	0.391	0.423	0.430	0.435	0.428	0.422	0.408	0.393	0.364	0.334	(40)	
	taud		2.468	2.397	2.306	2.232	2.260	2.292	2.311	2.345	2.364	2.390	2.412	2.484	(41)	
	Ebn at Noon		250	264	267	267	268	266	267	265	259	246	235	239	(42)	
	Edh at Noon		23	29	36	42	42	41	40	37	34	29	24	21	(43)	
(44) All-Sky Solar Radiation	RadAvg		487	751	1151	1559	1939	2211	2269	1958	1442	899	531	419	(44)	
	RadStd		53	93	156	129	162	118	115	108	70	99	57	49	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)		+0.52	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+115	N/A

Nomenclature: See separate page